Stochastic Processes – 2021/22 Semester 1

Tutorial 7

1. Let $\{N(t), t \geq 0\}$ be a (homogeneous) Poisson process. Let $0 < u_1 < u_2 < t$ be fixed real numbers.

   (a) i. For any $k = 0, 1, \ldots$ and for any $j = 0, \ldots, k$, find the conditional probability
   $$\mathbf{P}(N(u_1, u_2) = j \mid N(t) = k).$$
   ii. Find $\mathbf{Cov}(N(u_1, u_2), N(0, t))$.

   (b) Assume that passengers arrive at a train station in a Poisson process with rate $\lambda = 10$ (per minute). Given that there are 620 arrivals within the time interval $(2pm, 3pm)$, find the (conditional) probability that there were no arrivals between 2.29 and 2.31.

   (c) Assume that each passenger is female with probability 2/3 (and male otherwise) independently of everything else. Find the probability that, within the time interval $(2.29, 2.31)$, two females arrive, given that the total number of arrivals is at most 3. Does this probability depend on the length of the time interval?

2. Joe is a newspaper seller. His customers arrive according to a Poisson process with rate 1 per minute.

   (a) At any given time, what is the probability that at least five minutes have elapsed since his last customer has arrived?

   (b) Each customer, independently of all others, has probability α of being female.
   i. What is the distribution of the time (in minutes) between the arrivals of successive female customers?
   ii. In the case $\alpha = 0.2$, find the probability that more than two female customers arrive in a five-minute period, given that more than three male customers arrive in this period.

   (c) Joe talks to every third customer.
   i. What is the distribution of the time (in minutes) between Joe's successive conversations?
   ii. What is the probability that more than three minutes elapse between any one conversation and the next? [*Hint: note that this is the probability that, in the three minutes following the first conversation, no more than two further customers arrive.*]

3. Let $T_1, \ldots, T_k$ be independent random variables, exponentially distributed with respective means $\lambda_1^{-1}, \ldots, \lambda_k^{-1}$, and let the random variable $T = \min(T_1, \ldots, T_k)$.

   (a) By considering the relation between the exponential distribution and the Poisson process (i.e. $T_1, \ldots, T_k$ may be regarded as the times of occurrence of the first point in each of k independent Poisson processes on $\mathbb{R}_+$ with respective rates $\lambda_1, \ldots, \lambda_k$), and using the Superposition Theorem, show that T has an exponential distribution with mean λ^{-1} where $\lambda = \sum_{j=1}^{k} \lambda_j$.

   (b) Also show the above result directly.

4. Male motorcycle injuries occur as a Poisson process of rate 10 per week, while female motorcycle injuries occur as an *independent* Poisson process of rate 2 per week. (Clearly no motorcycle ever carries more than one person.) Consider both processes from a given time (0 say).

   (a) Find the probability that 8 male injuries occur before 4 female (*use the Superposition Theorem and consider the first 11 injuries*).

   (b) Let $X_0 = 0$ and let the random variable X_n be the number of male injuries *minus* the number of female injuries after a *total* of n injuries have occurred.

      i. Explain why the process $(X_n,\ n \geq 0)$ is a simple random walk and determine its transition probabilities.

      ii. Now let x be the probability that the total count of female injuries *ever* exceeds the total count of male injuries. Using the theory of simple random walks, determine x.

5. An insurance company issues n different types of policy. Claims, for all types of policy combined, arise as a Poisson process with intensity λ and the probability that any such claim which arises is under a policy of type k $(1 \leq k \leq n)$ is p_k (where $\sum p_k = 1$), independently of all other claims.

   (a) Explain why claims of type k arise as a Poisson process of intensity λp_k and these n processes are independent of each other.

   (b) Now suppose that $p_k = 1/n$ for all k.

      i. Starting from an arbitrary origin in time, find the mean and variance of the time until a total of n claims have occurred.

      ii. Find the mean and variance of the number of *different types* of policy on which claims arise in any time interval of length 1.

6. Fred collects stamps. There is a set of two, the 1d green and the 2d blue of 1896, that he particularly wants to obtain. Starting at time 0, examples of the 1d turn up as a Poisson process with rate λ_1 per year, while examples of the 2d turn up as an independent Poisson process with rate λ_2 per year. Find the expectation of the time (in years) required to obtain at least one example of each (*slightly tricky*). [*Hint: one possibility here is to use the Superposition Theorem, and condition on whether the 1d or the 2d turns up first.*]